PROGRAMMING FOR PROBLEM SOLVING

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Why Programming?

We solve problems every day.

But what happens when the problem becomes too large, too repetitive, or too complex for a human to handle?

The Big Question

Can we teach a computer how to solve the problem?

That's where Programming begins.

What is Programming?

Programming is the process of telling a computer how to solve a problem.

PROBLEM $\longrightarrow$ **INSTRUCTIONS** $\longrightarrow$ **COMPUTER** $\longrightarrow$ **RESULT**

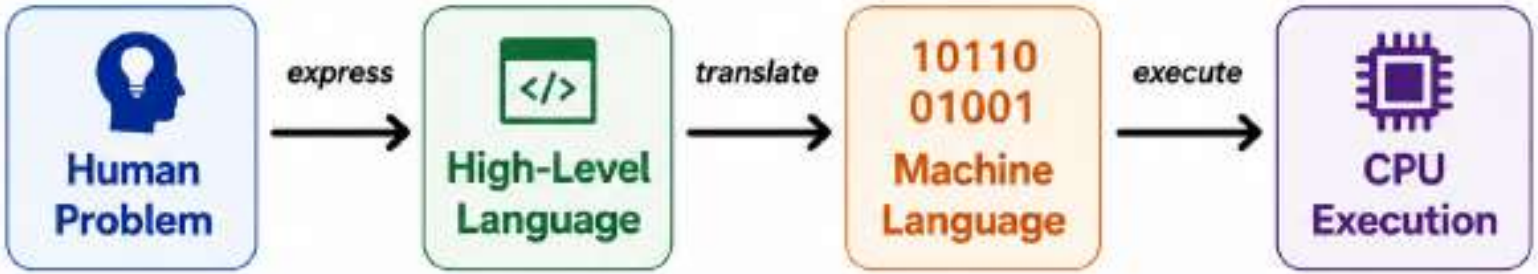
Think Like a Programmer

Understand the problem $\rightarrow$ **Break it into steps** $\rightarrow$ **Express the steps** $\rightarrow$ **Execute and test**

A program is a precise set of instructions that a computer can execute.

How Does a Computer Understand Our Instructions?

Humans think in ideas. Computers execute instructions.



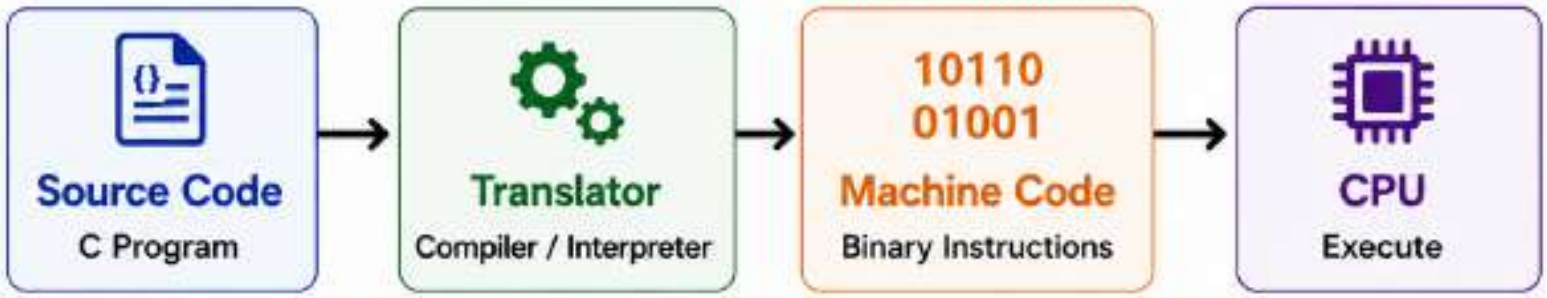
Three Levels of Programming Languages

- 101 010 • **Machine Language** -- Binary instructions - CPU
- MOV ADD • **Assembly Language** -- Mnemonics
- C / C++ Python • **High-Level Language** -- Human-readable languages

Key idea: A program written by humans must eventually become instructions that the computer can execute.

How Is a Program Translated?

We write programs for humans to understand.
The CPU needs machine instructions.



Compiler

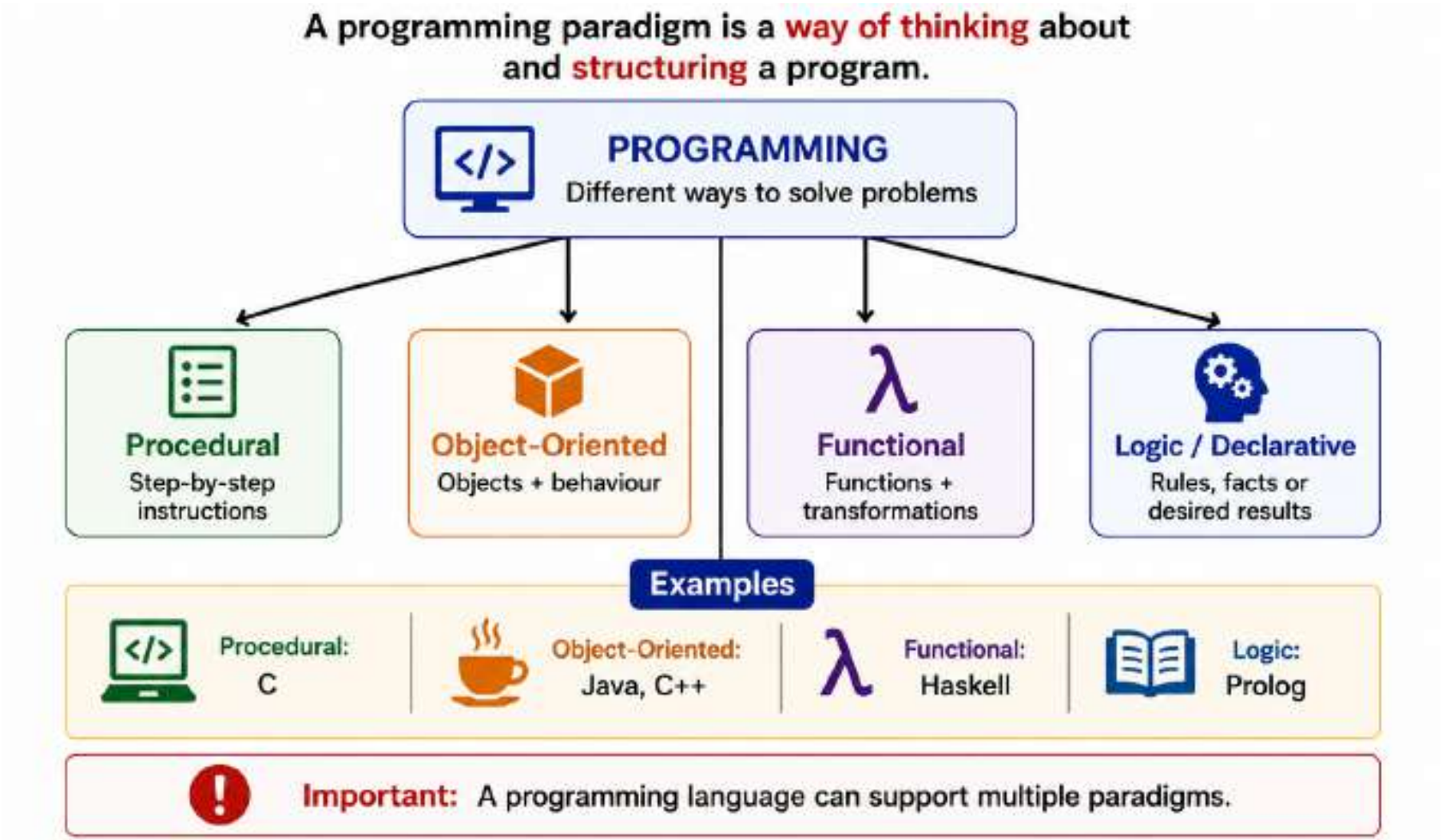
- Translates the program **before** execution
- Produces executable/object code
- C is typically **compiled**

Interpreter

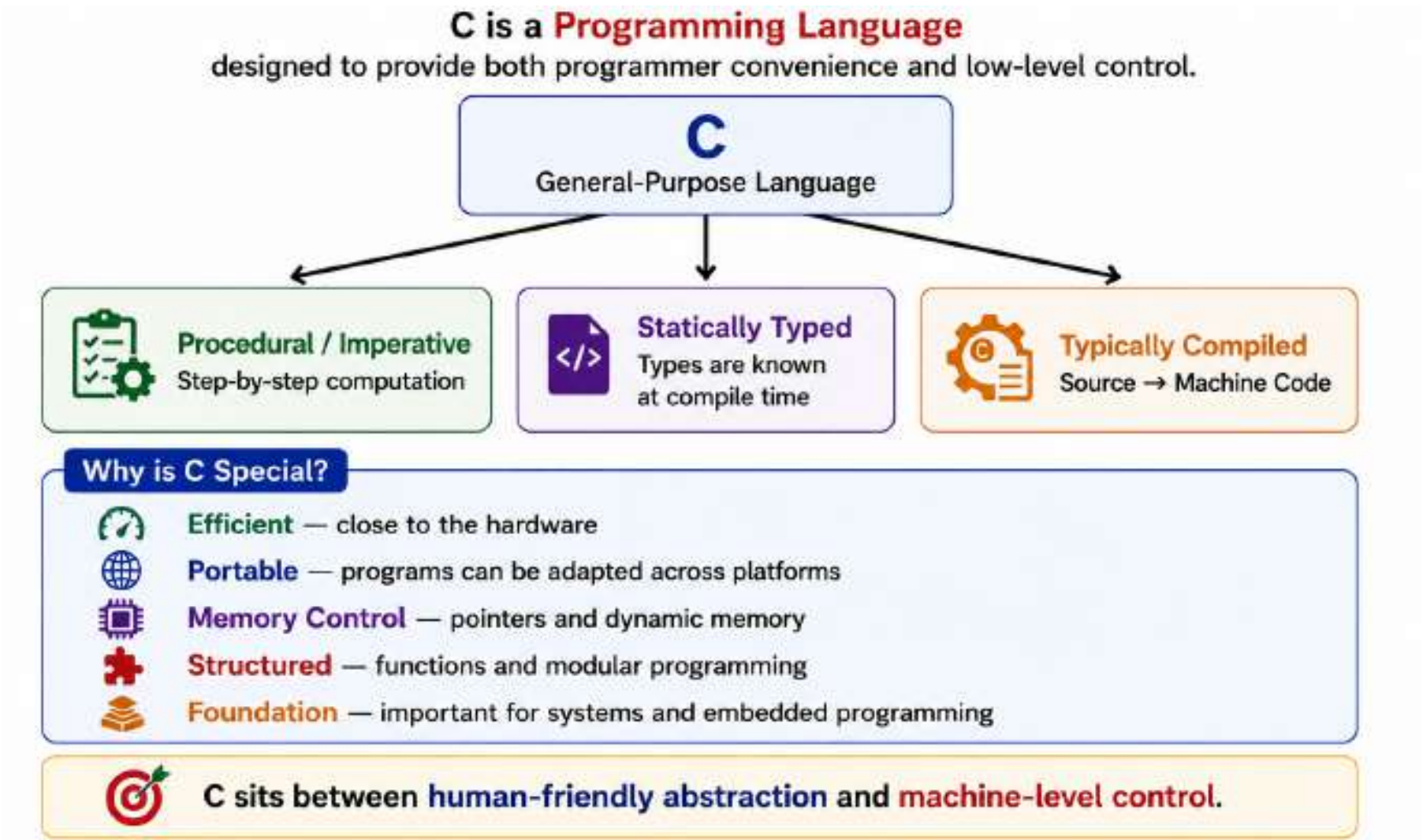
- Executes through a language **implementation**
- Translation and execution are closely **combined**
- Commonly associated with **Python**

Important: Compiler and interpreter describe *how programs are translated/executed* —they are **not programming paradigms.**

Programming Paradigms

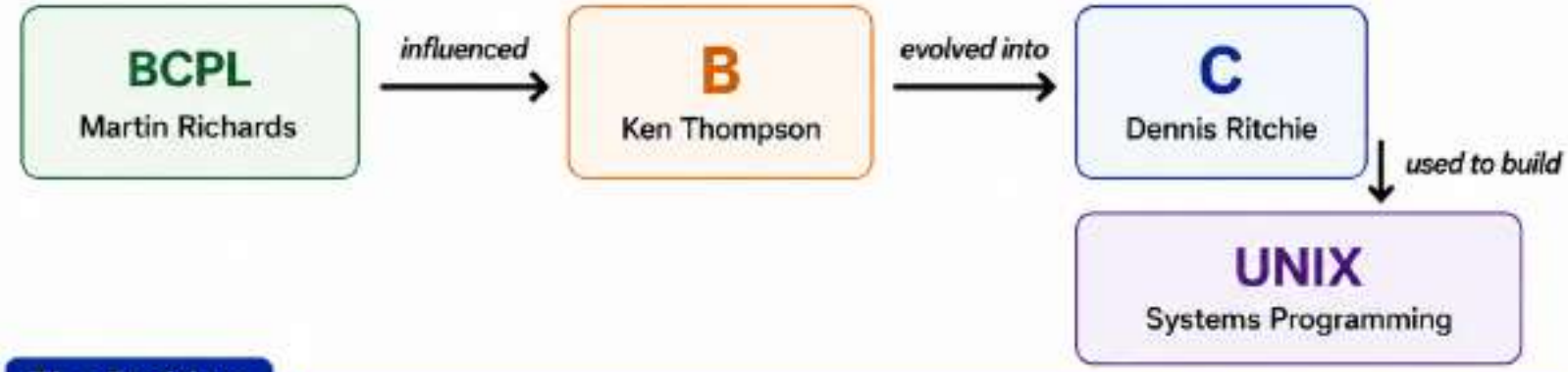


Where Does C Fit?



Why Was C Created?

C was born from a real problem:
How can we build powerful system software efficiently?



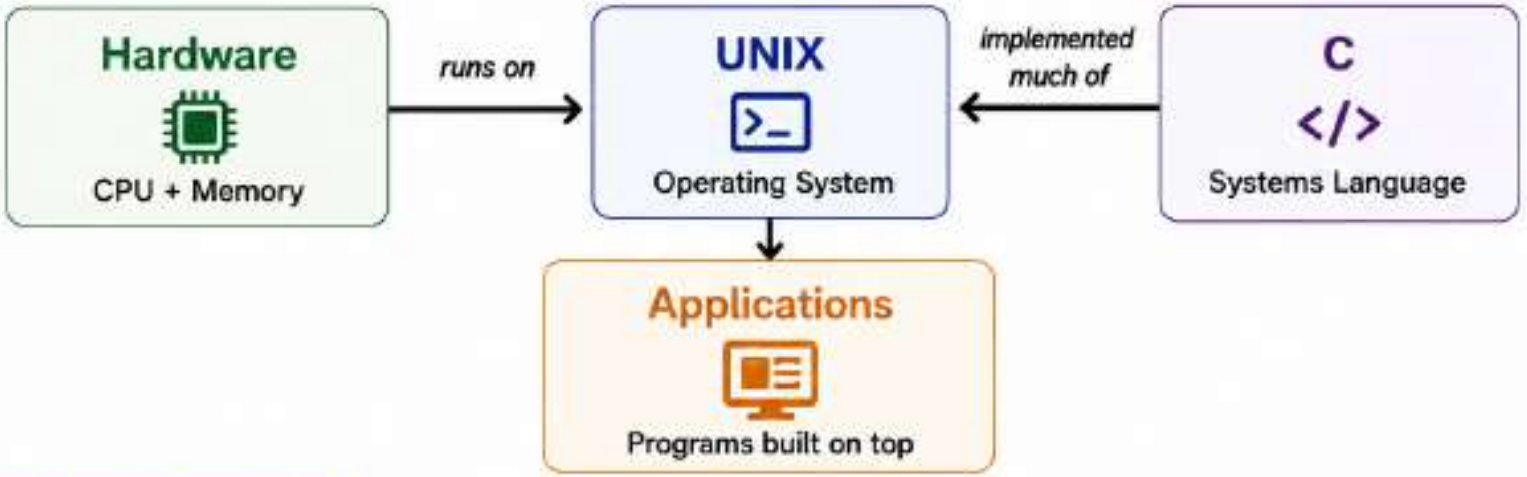
The Problem

- Assembly language offered **low-level control** but was difficult to develop and maintain.
- Higher-level languages offered **abstraction** but were not ideal for some systems programming tasks.
- C aimed to provide a practical balance between **abstraction, efficiency** and **hardware control.**

C was developed at **Bell Labs** during the **early 1970s.**

C and UNIX: A Turning Point in Computing

C became powerful because it was used to build real system software.

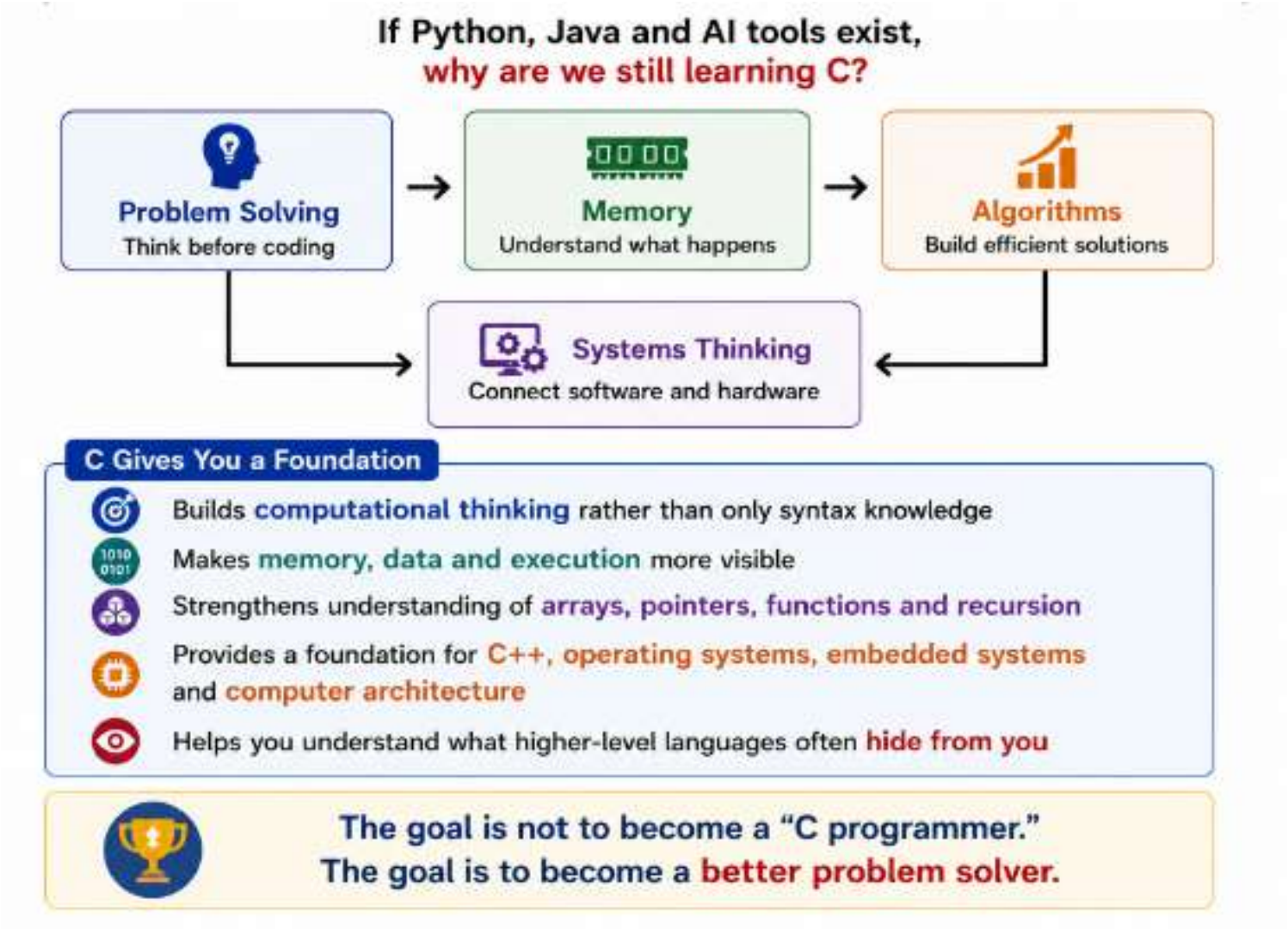


Why Did This Matter?

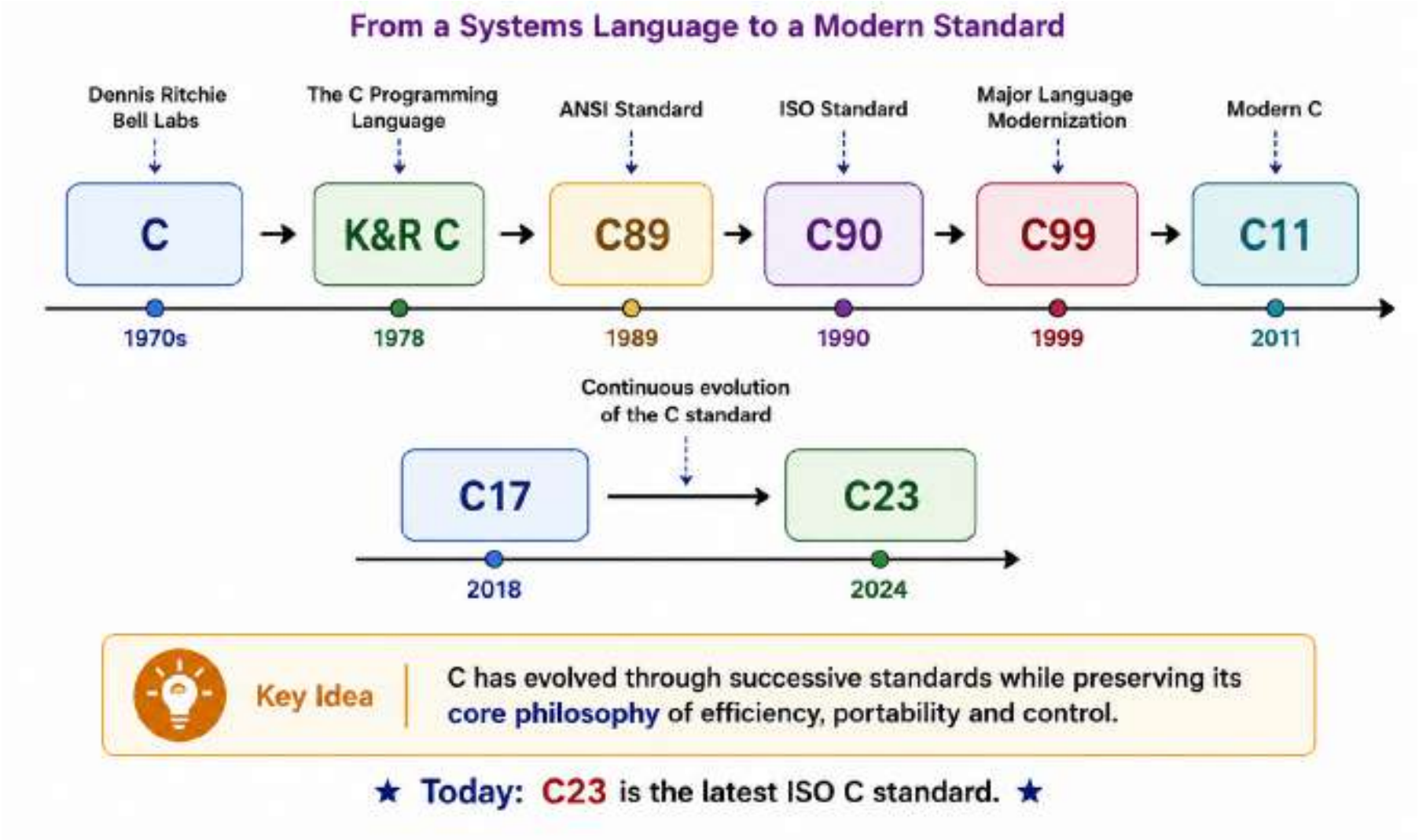
- C allowed large portions of system software to be written more conveniently than assembly.
- Programs written in C could be adapted to different computer architectures.
- The success and portability of UNIX helped C spread beyond its original environment.

C was no longer just a language experiment — it became a practical tool for building systems.

Why learn C in 2026?



Evolution of C?



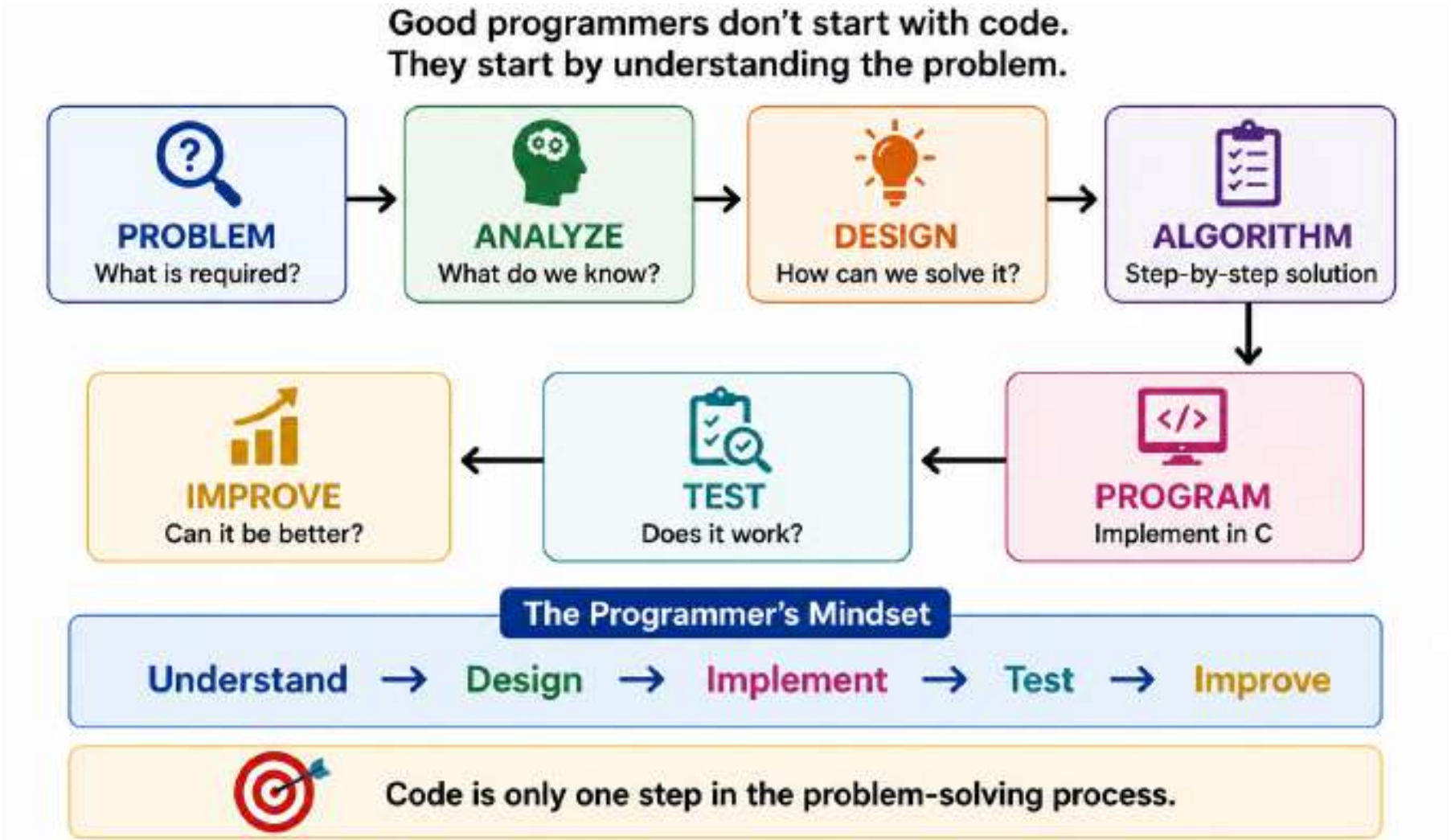
What changed in C?

Standard	Year	Key Highlights
C89	1989	First major ANSI standardization of C
C90	1990	ISO standardization of C
C99	1999	// comments, declarations in `for` loops, variable-length arrays, `inline`, improved integer types
C11	2011	Multithreading support, atomics, `_Generic`, improved Unicode support and safer library features
C17	2018	Clarifications, corrections and defect fixes; largely a refinement of C11
C23	2024	Further language modernization, including `nullptr`, `true/false` as keywords, digit separators and other improvements

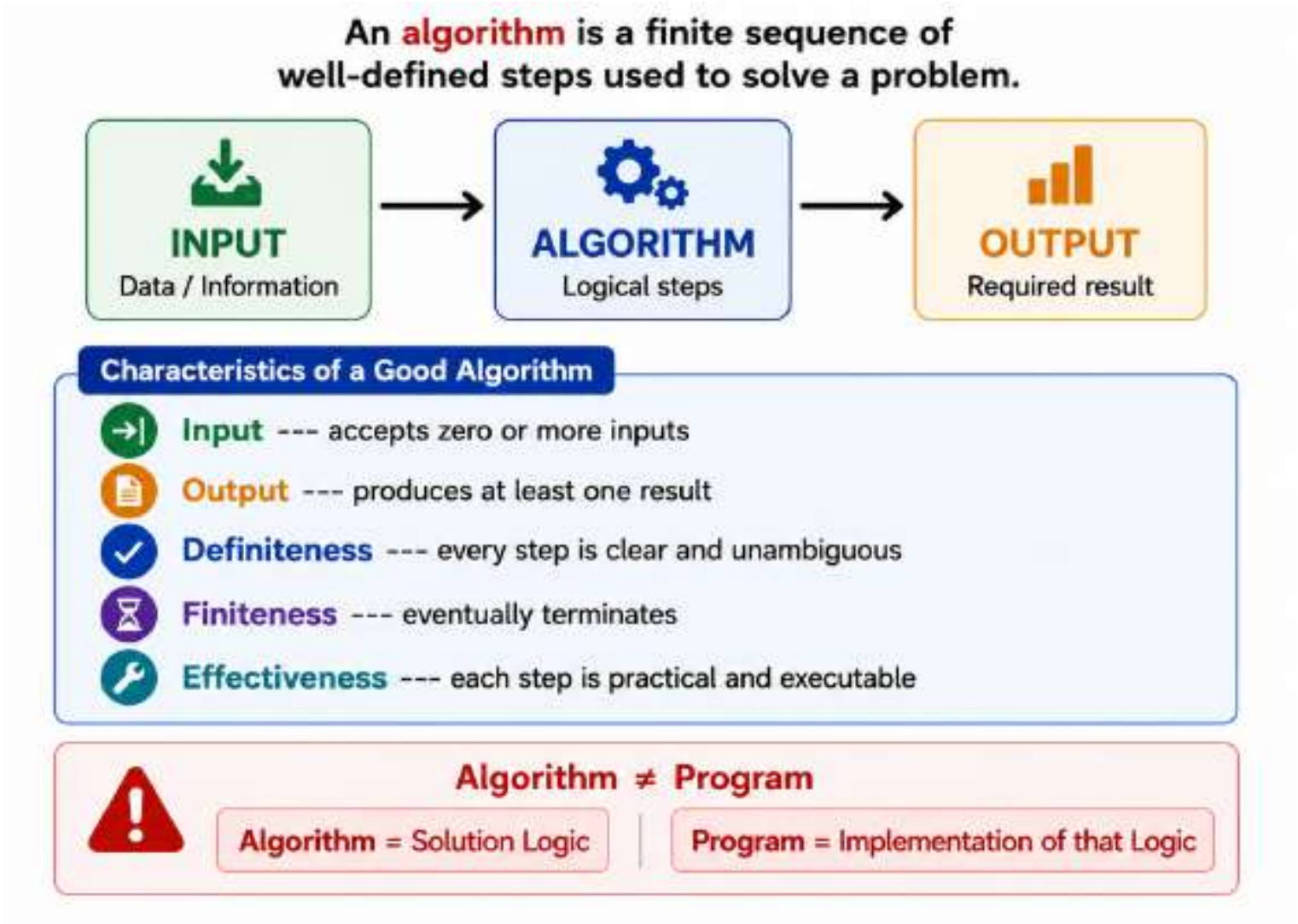
The Important Point

New standards add capabilities and improve the language, but C continues to emphasize efficiency, portability and control.

Before We Write Code: How Do We Solve a Problem¿



What is an Algorithm?



Algorithm Example: Finding the Largest of Three Numbers

Problem: Find the largest among three numbers

INPUT	PROCESS	OUTPUT
A, *B*, *C*	Compare the three values and identify the largest	*Largest*

Algorithm

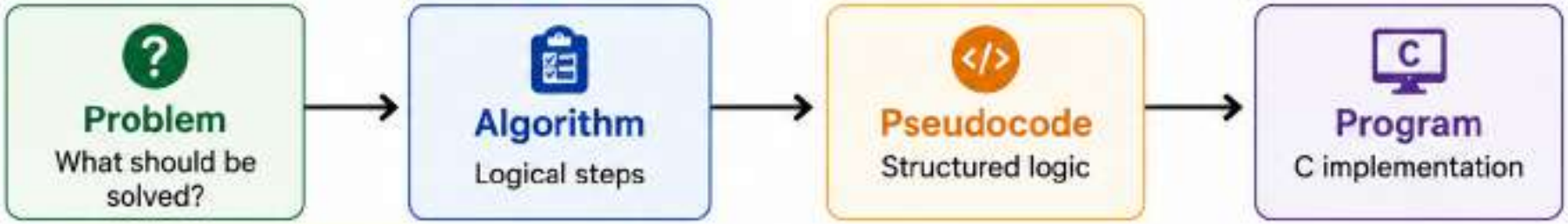
1. Read three numbers *A*, *B* and *C*.
2. Compare *A* with *B* and *C*.
3. If *A* is greater than both, set *Largest* = *A*.
4. Otherwise, compare *B* and *C*.
5. If *B* is greater than *C*, set *Largest* = *B*.
6. Otherwise, set *Largest* = *C*.
7. Display *Largest*.

Think Before You Code

We have solved the problem logically --- without writing C code.

What is Pseudocode?

Pseudocode is a structured way of describing an algorithm using simple, programming-like language.

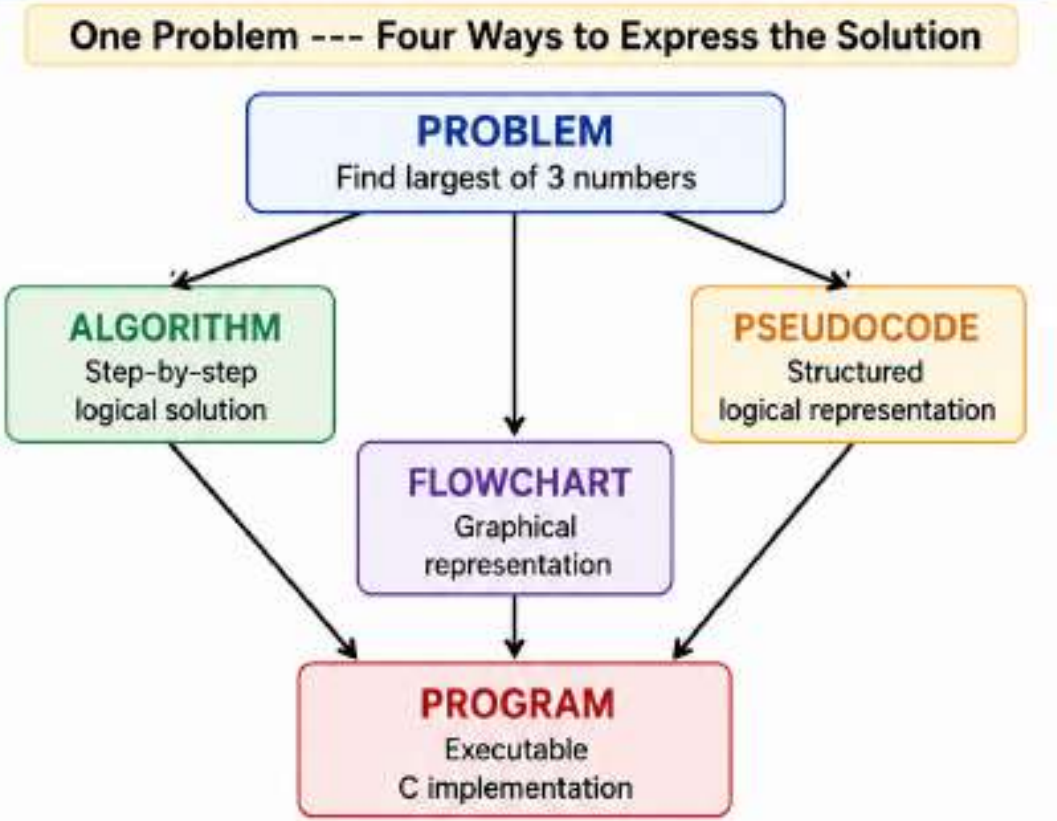


Example: Largest of Three Numbers

```
START
READ A, B, C

IF A > B AND A > C
   Largest <- A
ELSE IF B > C
   Largest <- B
ELSE
   Largest <- C

PRINT Largest
STOP
```

Pseudocode focuses on logic, not programming-language syntax.

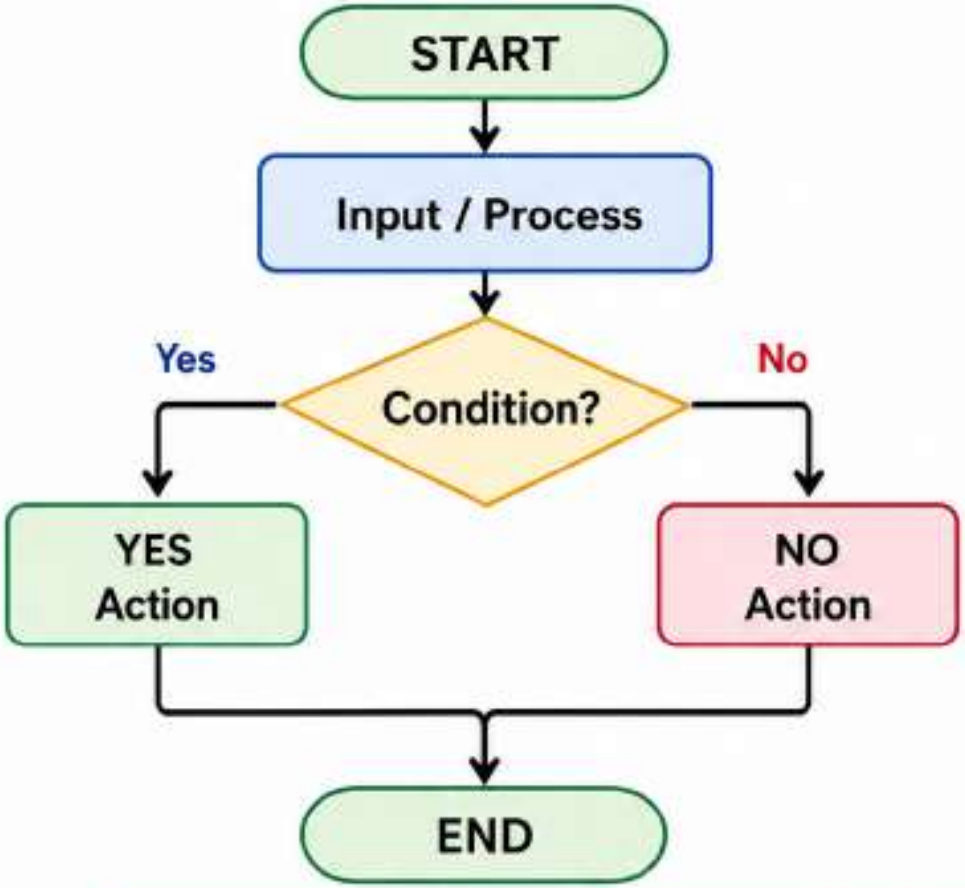
Algorithm Vs Pseudocode Vs Flowchart Vs Program



Representation	Focus	Language Dependent?
Algorithm	Logic / Steps	No
Pseudocode	Structured Logic	No
Flowchart	Visual Logic	No
C Program	Implementation	Yes

What is Flowchart?

A flowchart is a graphical representation of an algorithm.



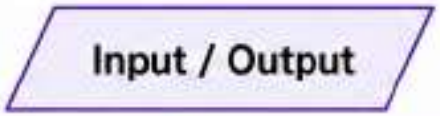
Why Use Flowcharts?

- **Visualize** the flow of a solution
- Make **decisions and branches** easy to see
- Help **identify missing or incorrect logic**
- **Communicate** a solution before writing code

Flowchart = Visual representation of the algorithm

Flowchart Symbol

Each symbol represents a specific type of operation.

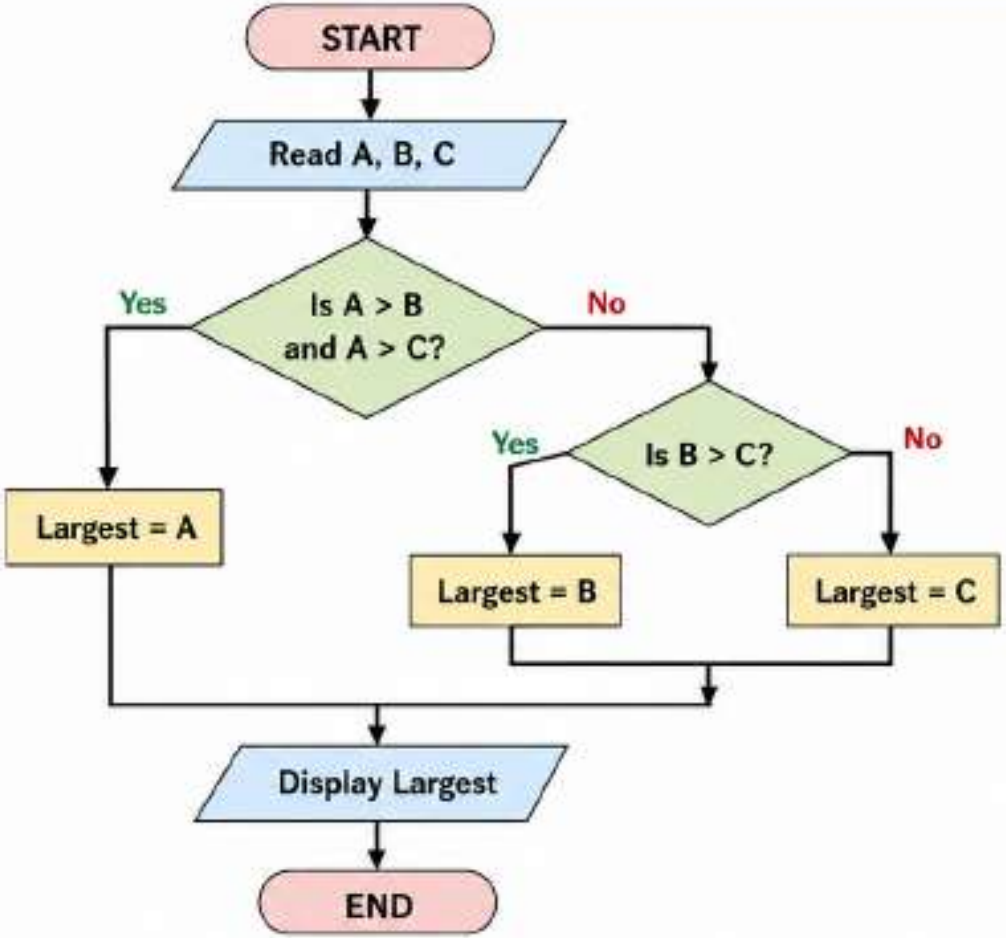
Symbol	Name	Purpose
	Terminal	Indicates the beginning or end of a flowchart
	Process	Represents a calculation or processing step
	Input / Output	Represents reading input or displaying output
	Decision	Represents a condition with alternative paths
	Flow Line	Shows the direction in which the algorithm proceeds

Remember

Shape gives meaning. Arrow gives direction.

Flowchart Example: Finding the Largest of Three Numbers

Problem: Find the largest among A, B and C.



Algorithm (Step-by-Step)

1. Read three numbers A, B and C.
2. If A is greater than both B and C, set Largest = A.
3. Otherwise, if B is greater than C, set Largest = B.
4. Otherwise, set Largest = C.
5. Display Largest.

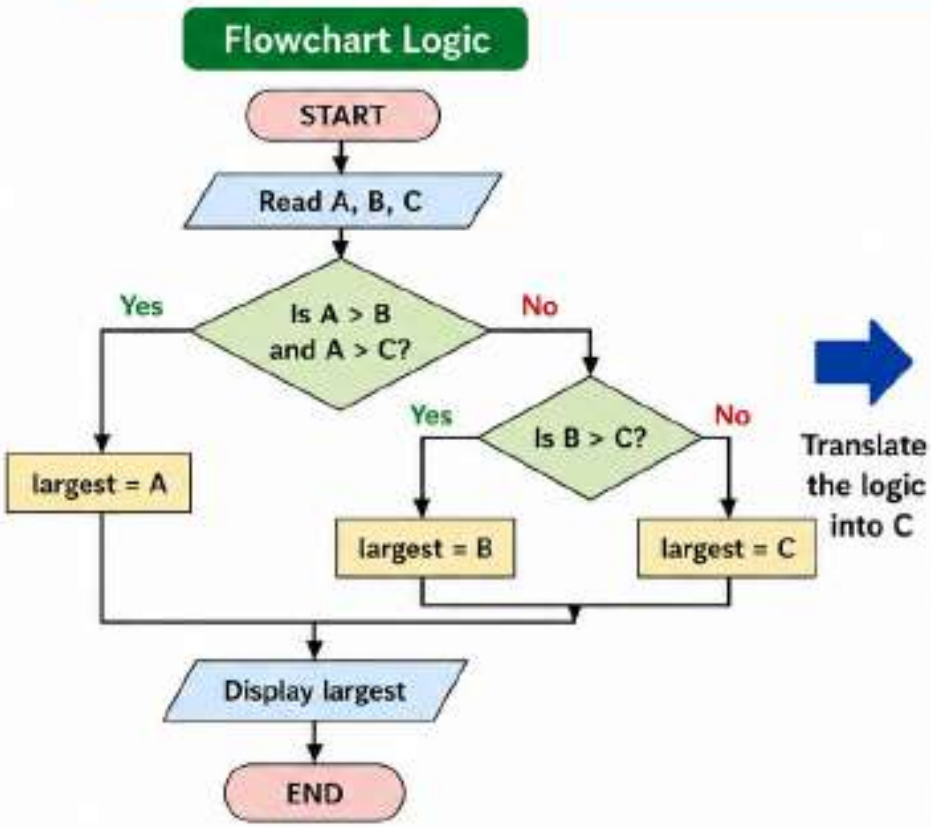
Legend (Flowchart Symbols)

Oval	: Start / End	
Parallelogram	: Input / Output	
Rectangle	: Process / Action	
Diamond	: Decision / Condition	
Arrow	: Flow Direction	

Note: The flowchart represents the logic. The same logic will be written in C in the next step.

From Flowchart to C Program

The logic stays the same – only the representation changes.



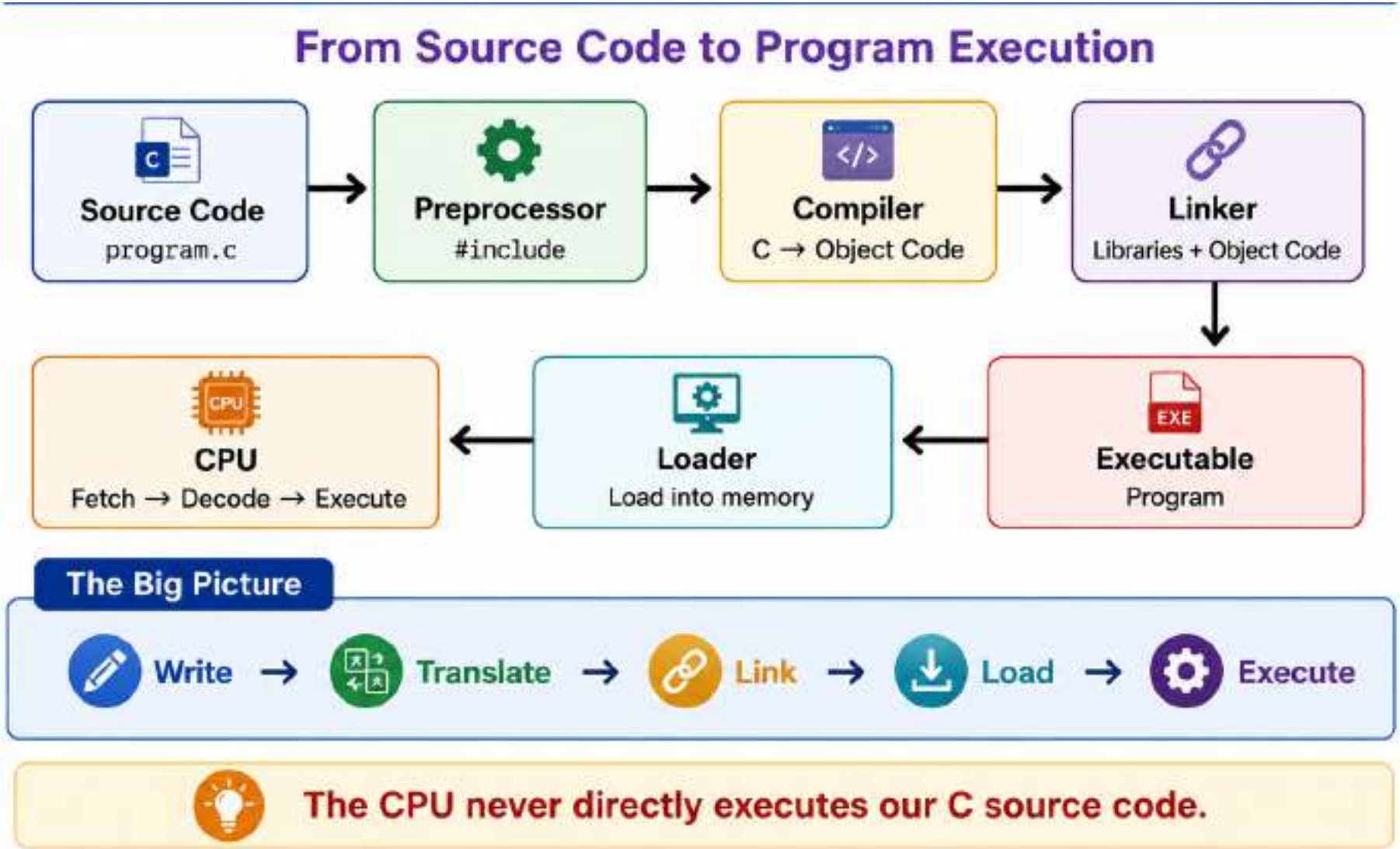
Translate the logic into C

C Program

```c
#include <stdio.h>

int main()
{
    int A, B, C, largest;       // Variable declaration

    scanf("%d %d %d", &A, &B, &C);  // Read input

    if (A > B && A > C)
        largest = A;                // A is largest
    else if (B > C)
        largest = B;                // B is largest
    else
        largest = C;                // C is largest

    printf("Largest = %d", largest);  // Display result

    return 0;
}
```

Problem – What to solve? → **Algorithm** – Step-by-step logic → **Pseudocode** – Structured steps → **Flowchart** – Visual representation → **C Program** – Executable code

What happens When We Run a C Program?



What Will We Learn in Programming for Problem Solving?

From Basic Programming to Real-World Data Processing

UNIT 1
C Fundamentals

- Data Types
- Variables
- Operators
- Expressions
- Decisions
- Boolean Logic

→

UNIT 2
Functions & Pointers

- Problem Decomposition
- Functions
- Arguments
- Pass-by-Value
- Pass-by-Reference
- Pointers

→

UNIT 3
Data Collections

- Arrays
- Strings
- Searching
- Linear Search
- Binary Search
- Sorting

→

UNIT 4
Iteration & Recursion

- Loops
- Break / Continue
- Recursion
- Iteration vs Recursion
- Memory Usage
- Dynamic Memory

→

UNIT 5
Structures & Files

- Structures
- Structure Functions
- Dynamic Memory
- File Handling
- Records
- Text / Line Processing

The Big Journey

Think → Design → Code → Organize Data → Solve Complex Problems

Unit 1 -- C Fundamentals & Decision Making

From Problem Understanding to Basic C Programs

What We Will Learn

- **Data Types and Variables** — `int`, `float`, `char`, constants
- **Operators and Expressions** — arithmetic, relational, logical and assignment operators
- **Assignment and Expressions** — evaluate and trace expressions
- **Decision Making** — `if`, `else if`, `else`
- **Boolean Expressions** — conditions and logical decision making
- **Problem Solving** — convert simple problems into C programs

Guided Practice

Variable tracing → Expressions → Conditions → Decision-making programs

Unit 2 -- Functions, Decomposition & Pointers

From One Large Program to Modular Solutions

What We Will Learn

- **Problem Decomposition** — break a problem into smaller sub-problems
- **Functions** — declaration, definition, calling and return values
- **Function Arguments** — parameters and different ways of passing values
- **Pass-by-Value and Pass-by-Reference** — understand how data is passed to functions
- **Pointers** — addresses, pointer variables and memory access
- **Modular Programming** — design reusable functions for complex problems

Guided Practice

Decompose → Design Functions → Pass Data → Build Modular Programs

Unit 3 -- Arrays, Strings, Searching & Sorting

From Single Values to Collections of Data

What We Will Learn

- **Arrays** — declaration, initialization, indexing and traversal
- **Array Operations** — reading, modifying and processing elements
- **Strings** — character arrays and string manipulation
- **String Functions** — `strlen()`, `strcpy()`, `strcat()`, `strcmp()`, `strchr()`, `strstr()`
- **Searching** — Linear Search and Binary Search
- **Sorting** — Bubble, Selection, Insertion, Quick and Merge Sort

Guided Practice

Store → Traverse → Search → Sort → Process Data

Unit 4 -- Iteration, Recursion & Dynamic Memory

Repeating Solutions and Managing Memory

What We Will Learn

- **Loops** — `for, while and do-while`
- **Iteration** — solve repeated problems systematically
- **Break and Continue** — control loop execution
- **Recursion** — recursive functions and base cases
- **Recursion vs Iteration** — compare solutions and memory behaviour
- **Dynamic Memory Allocation** — `malloc(), calloc(), realloc() and free()`

Guided Practice

Unit 5 -- Structures, Dynamic Data & File Handling

From Simple Data to Real-World Records

What We Will Learn

- **Structures** — group different types of data into a single record
- **Structure Variables** — create, access and modify structured data
- **Functions with Structures** — pass structures to functions
- **Dynamic Memory with Structures** — manage unpredictable or large inputs
- **File Handling** — create, open, read, write and close files
- **File Processing** — process records, text, characters, words and lines

Guided Practice